

Text-Only Detection of AI-Generated RedNote Posts: A TF-IDF and Transformer Comparison

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Abstract

This report summarizes the implemented RedNoteC project for binary detection of AI-generated Chinese RedNote/Xiaohongshu posts. The intended task is to compare lightweight and transformer-based NLP methods for Chinese social media AI-text detection on the supervised RedNote-Vibe human and AI-generated training files [Li et al., 2026]. Our inputs are built only from post title and content. Metadata is retained for auditing and subgroup reporting but is not used as model input. We compare character n-gram TF-IDF with logistic regression against a fine-tuned Chinese RoBERTa-style classifier. On the fixed test split of 8,833 posts, the transformer obtains higher macro-F1 (0.9615 vs. 0.9457), AUROC (0.9964 vs. 0.9899), average precision (0.9882 vs. 0.9648), and AI recall (0.9724 vs. 0.8959). The TF-IDF baseline has slightly higher AI precision at the default 0.5 threshold (0.9136 vs. 0.8964) and fewer human false positives. These results support a cautious conclusion: the transformer improves recall-oriented detection on this split, but the baseline remains strong and substantially simpler.

1 Introduction

AI-generated Chinese social media detection is a practical problem for platforms such as Xiaohongshu/RedNote, where informal style, slang, emojis, hashtags, and mixed-language usage complicate direct transfer from formal or English-focused AI-text detection settings. Recent Chinese benchmark work makes the same broader point: Chinese AI-generated text detection is affected by language-specific segmentation, idioms, informal abbreviations, domain variation, and generator variation [Qing et al., 2026]. RedNote-Vibe is directly relevant because it is an arXiv dataset preprint fo-

cused on RedNote/Xiaohongshu lifestyle social media, providing human-written posts, synthetic AI-generated variants, metadata, and a later unlabeled exploration set [Li et al., 2026].

Prior work on AI-generated text detection motivates comparing lightweight lexical methods with contextual neural models. Statistical and feature-based approaches are attractive because generated text can leave detectable distributional traces in word choice, repetition, and phrase usage; for example, GLTR showed that statistical signals can help expose generation artifacts and support human detection [Gehrmann et al., 2019]. TF-IDF features combined with linear classifiers therefore provide a transparent and reproducible baseline for text classification [Salton and Buckley, 1988, Pedregosa et al., 2011]. However, Chinese social media text adds extra difficulty because surface-level features are affected by word segmentation, short informal posts, emojis, hashtags, and platform-specific expressions [Qing et al., 2026, Li et al., 2026]. Contextual encoders such as BERT and RoBERTa are useful to test because they model words in context rather than treating terms as independent features [Devlin et al., 2019, Liu et al., 2020]. For Chinese specifically, whole-word masking has also been proposed to better represent multi-character word units in BERT-style models [Cui et al., 2019]. This makes a controlled comparison between TF-IDF and Chinese RoBERTa a suitable first step for evaluating AI-generated RedNote post detection from text alone.

The central research question is: for detecting AI-generated RedNote posts from title and content alone, does a contextual transformer provide a meaningful advantage over a lightweight TF-IDF baseline? This comparison is useful because TF-IDF and linear classifiers remain strong and re-

producibile text-classification baselines [Salton and Buckley, 1988, Pedregosa et al., 2011], while BERT/Roberta-style encoders provide contextual representations at much higher computational cost [Devlin et al., 2019, Liu et al., 2020].

Moreover, this research is not only a comparison of two model families. It is also a question about what kind of evidence is useful for a practical AI-text detector. In an imbalanced setting where human posts are much more common than AI-generated posts, a high accuracy score can hide the detector’s real weakness. A model may classify most human posts correctly and still miss many AI-generated examples. For this reason, the project treats accuracy as only one part of the evaluation. AI recall, false negatives, AUROC, average precision, subgroup behavior, and training cost are all necessary for judging whether the transformer is meaningfully better than the baseline.

The comparison is also designed to avoid an unfair advantage from metadata. RedNote-Vibe includes useful metadata such as domain, generator family, generator model, timestamp, and engagement-related fields. These fields are valuable for auditing and subgroup analysis, but using them as direct model inputs could create shortcut learning. For example, if a model learned that a particular generator name or metadata pattern was associated with AI text, the result would not reflect genuine text-only detection ability. Therefore, the main experiment deliberately uses only title and content as classifier input. This makes the evaluation narrower, but also cleaner: the measured performance reflects what can be learned from the language of the post itself.

2 Methodology

2.1 Dataset and Input Policy

The supervised data comes from RedNote-Vibe [Li et al., 2026]. The repository loads 51,878 human rows and 7,254 AI rows, with no malformed or missing-required rows recorded in the manifest. After conservative cleaning and exact-text deduplication, 58,882 rows remain. The labels are defined as 0 for human and 1 for AI.

The main non-negotiable design constraint is text-only training. The processed model input is built from title and content:

```
标题: {note_title}
正文: {note_content}
```

If only content is present, only the content form is used; title-only rows are permitted by configuration, although our cleaning report records zero kept title-only rows. Metadata fields such as domain, model_family, model, timestamps, and engagement counts are retained for auditing and subgroup reports only. The training loader selects only text and label.

Cleaning is intentionally light: whitespace is normalized, carriage returns are standardized, repeated blank lines are collapsed, and exact duplicate texts are removed. Emojis, hashtags, punctuation, informal wording, and code-switching are preserved. This is consistent with the focus on social-media language, while avoiding unimplemented stronger normalization choices such as jieba segmentation or stop-word removal.

The decision to preserve emojis, hashtags, punctuation, slang, and mixed-language expressions is important for this task. In many formal NLP tasks, such elements may be treated as noise. In Chinese social-media AI-text detection, however, they can carry stylistic information. Human RedNote posts often use platform-specific formatting, informal emotional markers, fragmented sentences, and visual expression through emojis or punctuation. AI-generated posts may imitate these patterns, but may still differ in consistency, repetition, or phrasing. Removing these signals through aggressive preprocessing would make the input cleaner, but it could also remove precisely the evidence that distinguishes human and AI writing.

At the same time, conservative cleaning has a limitation. It means that the models may partly learn surface style rather than deeper semantic differences. This is especially relevant for the TF-IDF baseline, which depends on character n-grams. If the dataset contains generator-specific repeated phrases, templates, or formatting habits, a character n-gram model can capture them very effectively.

This is useful for detection on the fixed split, but it may not fully generalize to future AI-generated posts if newer generators produce more diverse or human-like text. Therefore, the cleaning policy supports a realistic social-media input setting, but the results should still be interpreted as supervised in-distribution evidence rather than proof of long-term robustness.

Split	Rows	Human	AI
Train	41,217	36,140	5,077
Validation	8,832	7,741	1,091
Test	8,833	7,747	1,086

Table 1: Fixed processed split. The split uses seed 42 and label+domain stratification.

2.2 Splitting

All models use the same 70/15/15 train/validation/test split (Table 1). Seed 42, random stratified splitting, and stratification by label plus domain are used. The split code validates that exact duplicate text values do not cross split boundaries. Group-aware splitting is not used because the candidate `note_id` column is not sufficient for reconstructing related human/AI groups in this processed data.

2.3 Models

The TF-IDF baseline is implemented as a scikit-learn pipeline [Pedregosa et al., 2011]: `TfidfVectorizer` followed by class-weighted logistic regression. It uses character n-grams from length 2 to 5, `min_df=2`, `max_df=0.95`, sublinear term frequencies, and a maximum of 300,000 features. Character n-grams are a conservative choice for Chinese social-media text because they avoid committing to a word segmenter while preserving local surface patterns. The saved model reaches 300,000 features and has 300,001 logistic-regression parameters including the intercept.

The transformer model is implemented with Hugging Face Transformers [Wolf et al., 2020]. The repository requested and loaded `hfl/chinese-roberta-wwm-ext`. This model

is a Chinese BERT/RobERTa-style whole-word-masking encoder, related to BERT and RoBERTa pretraining methods [Devlin et al., 2019, Liu et al., 2020, Cui et al., 2019]. The saved configuration has 12 layers, hidden size 768, 12 attention heads, intermediate size 3,072, and 102,269,186 trainable parameters. Fine-tuning uses maximum length 256, batch size 4, gradient accumulation 2, learning rate 2×10^{-5} , one epoch, weight decay 0.01, warmup ratio 0.06, and validation macro-F1 for best-model selection.

Model	Implemented configuration
TF-IDF + LR	Character 2–5 grams; 300k max features; class-weighted logistic regression; solver <code>liblinear</code> .
Chinese RoBERTa	Chinese RoBERTa WWM; max length 256; one epoch; effective batch size 8; 102.3M parameters.

Table 2: Implemented model comparison. Both models use only `text` as input and `label` as target.

The two models represent different assumptions about the detection problem. The TF-IDF model assumes that local lexical and character-level patterns are sufficient for classification. This is a reasonable assumption for Chinese social media because character n-grams can capture short phrases, punctuation habits, repeated expressions, and informal written patterns without requiring word segmentation. It is also a strong baseline because it is fast to train, easy to reproduce, and easier to inspect than a neural model. If this model performs well, it suggests that many AI-generated RedNote posts contain detectable surface-level regularities.

The transformer model makes a different assumption. Instead of treating character n-grams as independent features, it builds contextual representations of the whole input sequence. This can help when the difference between human and AI writing is not limited to repeated phrases, but appears in sentence flow, local coherence, topic development, or the relationship between title and content. However, the transformer also introduces higher computational cost and lower transparency. Its improvement should therefore be judged by whether it solves important errors that the baseline misses, rather than

by whether it gives a small increase in accuracy.

This distinction is central to the project. A transformer is not automatically the better practical choice just because it is more advanced. If the TF-IDF model were nearly equal on recall, calibration, and subgroup behavior, then the simpler model would be more attractive. Conversely, if the transformer substantially reduces missed AI posts or performs better on difficult subgroups, then its additional cost may be justified. The model comparison is therefore framed as a cost-performance and error-profile comparison, not only a leaderboard-style accuracy comparison.

2.4 Evaluation

Evaluation is shared across models. Each saved model scores the fixed test split, converts AI scores to labels using threshold 0.5, and writes metrics, predictions, subgroup reports, error CSVs, and diagnostic plots. The reported metrics include accuracy, balanced accuracy, per-class precision/recall/F1, macro-F1, weighted F1, log loss, AUROC, average precision, and confusion matrix. AUROC and average precision are computed from the AI-class score rather than from hard labels.

The evaluation metrics are selected to reflect the asymmetric nature of the detection task. False negatives and false positives have different practical meanings. A false negative means an AI-generated post is classified as human, so the detector fails to catch the content it is designed to identify. A false positive means a human post is incorrectly flagged as AI, which can be harmful if the detector is used for moderation or user-facing decisions. Because both errors matter, the report considers AI recall and AI precision together instead of treating one metric as sufficient.

Macro-F1 is also important because it gives equal weight to the human and AI classes. In this dataset, the human class is much larger than the AI class, so weighted F1 and accuracy are strongly influenced by human-post performance. Macro-F1 is more sensitive to whether the AI class is handled well. AUROC and average precision add another perspective because they evaluate ranking quality from model scores rather than only the fixed 0.5 decision thresh-

old. This matters because a practical detector may tune the threshold depending on whether the goal is high recall or high precision.

Log loss is included because it gives information about score confidence. Two models may produce the same hard-label prediction but differ in how confident or calibrated their probability estimates are. A lower log loss suggests that the model assigns better probabilities to the correct class, which can be useful if the detector is used as part of a larger review system rather than as a simple yes/no classifier. Therefore, the evaluation combines threshold-based metrics, ranking metrics, and probability-sensitive metrics.

3 Experiments

3.1 Overall Results

Table 3 reports the test results. The transformer improves most aggregate and recall-oriented metrics. In particular, AI recall rises from 0.8959 to 0.9724, reducing missed AI posts from 113 to 30. However, AI precision is lower for the transformer at the fixed 0.5 threshold, and human false positives increase from 92 to 122 (Table 4). This is an important trade-off rather than a uniformly better outcome.

The runtime evidence is also asymmetric. The transformer trainer log records 5,153 final steps and `train_runtime=4339.4742` seconds, which is approximately 72.3 minutes, or about 1 hour and 10 minutes. By contrast, the TF-IDF baseline took about 10 minutes to train.

Metric	TF-IDF + LR	Chinese RoBERTa
Accuracy	0.9768	0.9828
Balanced accuracy	0.9420	0.9783
Human precision	0.9855	0.9961
Human recall	0.9881	0.9843
AI precision	0.9136	0.8964
AI recall	0.8959	0.9724
AI F1	0.9047	0.9329
Macro-F1	0.9457	0.9615
Weighted F1	0.9767	0.9831
Log loss	0.1628	0.0774
AUROC	0.9899	0.9964
Average precision	0.9648	0.9882

Table 3: Test results on 8,833 posts. Metrics are from saved evaluation outputs.

Model	True	Pred. H	Pred. AI
TF-IDF	Human	7,655	92
	AI	113	973
RoBERTa	Human	7,625	122
	AI	30	1,056

Table 4: Confusion matrices on the test split. Rows are true labels and columns are predicted labels.

The main result is that the transformer advantage is more meaningful in recall-oriented metrics than in raw accuracy. Accuracy increases from 0.9768 to 0.9828, which is numerically small. If accuracy were the only metric, the transformer improvement might appear minor.

However, the confusion matrix shows a more important change: missed AI posts decrease from 113 to 30. This means the transformer catches many AI-generated posts that the TF-IDF model leaves undetected. For an AI-content detection task, this reduction in false negatives is one of the strongest arguments for using the transformer.

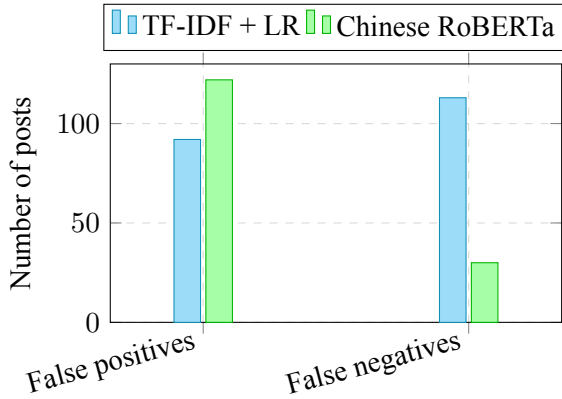


Figure 1: False-positive and false-negative errors on the test split. False positives are human posts predicted as AI; false negatives are AI posts predicted as human.

The tradeoff is that the transformer increases human false positives from 92 to 122. This is not a trivial issue. In a real deployment, incorrectly labeling human writing as AI could reduce user trust and create fairness concerns, especially if certain writing styles are more likely to be flagged. Therefore, the transformer should not be described as uniformly better. It is better for recall-oriented detection, but not necessarily better if the application prioritizes

avoiding false accusations against human users. A practical system would likely need threshold tuning, human review, or different operating points for different use cases.

The AUROC and average precision results suggest that the transformer not only performs better at the default threshold, but also ranks examples more effectively. This is important because the default threshold of 0.5 is not necessarily the best operational threshold. A moderation system could choose a higher threshold to reduce false positives, or a lower threshold to increase recall. Since the transformer has stronger ranking metrics, it may provide more flexibility for threshold selection. However, this claim should be tested explicitly on the validation set before being used as a deployment recommendation.

3.2 Subgroup Results

Subgroup results are available by domain, AI generator family, AI model, and text-length bucket. These are descriptive analyses of the fixed test split, not separate held-out-generator or held-out-domain experiments. For AI-only generator-family subgroups, AUROC is undefined because only positive examples are present; AI recall is the most interpretable number.

Domain variation. Subgroup results are available by domain, AI generator family, AI model, and text-length bucket. These are descriptive analyses of the fixed test split, not separate held-out-generator or held-out-domain experiments. For AI-only generator-family subgroups, AUROC is undefined because only positive examples are present; AI recall is the most interpretable number.

Domain subgroup results show that the transformer generally improves AI recall across domains (Table 5). The weakest domain for both models is 学习 (education/study): TF-IDF AI recall is 0.6944, while transformer AI recall is 0.9444. The transformer still has its lowest domain macro-F1 in this group (0.9139), indicating that the improvement does not remove all domain variation.

Domain	Rows	AI	TF-IDF R	RoBERTa R
健康	1,584	169	0.8817	0.9763
美食	1,342	141	0.8865	0.9716
情感	1,189	165	0.8909	0.9515
旅行	1,045	125	0.9040	0.9920
职场	940	126	0.9127	0.9603
穿搭	932	171	0.9649	0.9825
心理	735	114	0.8860	0.9825
学习	598	36	0.6944	0.9444
宠物	278	25	0.84	0.96
运动	190	14	0.8571	1.0

Table 5: Selected domain subgroup AI recall (R) on the test split.

The domain subgroup results are important because they show that overall metrics do not fully describe model behavior. The education/study domain is the clearest example. TF-IDF AI recall is much lower in this domain than in most others, while the transformer recovers a large portion of those missed AI posts. One possible interpretation is that study-related posts may contain more formal explanation, advice, or structured content, which can make AI-generated and human-written posts harder to separate using only local character patterns. The transformer may benefit from modeling broader contextual cues in these cases.

However, these subgroup results should be interpreted carefully. Some domains have far fewer AI examples than others, such as sports and pets. A very high recall value in a small subgroup may be unstable because a small number of additional errors would change the metric substantially. For this reason, subgroup performance should be treated as diagnostic evidence rather than definitive proof that the transformer is robust in every domain. The main value of the subgroup analysis is that it identifies where the baseline is weaker and where future evaluation should focus.

Generator-family variation. The generator-family results provide a similar diagnostic signal. The GLM family is the hardest group for both models, but the transformer still improves recall substantially. TF-IDF recall ranges from 0.7177 on glm to 0.9669 on gemini; transformer recall ranges from 0.8871 on glm to 1.0000 on qwen. This suggests that not all AI-generated posts are equally

easy to detect. Some generators may produce text with stronger surface regularities, while others may produce outputs that are less recognizable to a character n-gram model.

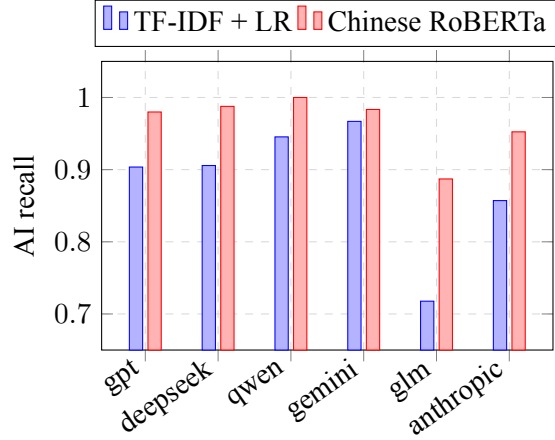


Figure 2: AI recall by generator family on the test split.

Because the generator-family analysis is computed only over AI-labeled rows from the same random test split, it should not be interpreted as a true unseen-generator generalization result. A stronger future experiment would hold out one generator family during training and test whether the model can generalize to it.

Length-bucket variation. Length-bucket results show a clearer pattern for TF-IDF: AI recall decreases from 0.9400 on short posts to 0.8247 on long posts. The transformer maintains higher recall across short, medium, and long buckets (0.9794, 0.9638, and 0.9691), but its AI precision is lowest on long posts. This suggests that threshold selection may matter if false positives on long human posts are costly.

The length-bucket results show another important limitation of surface-level detection. TF-IDF recall decreases as posts become longer, while the transformer maintains high recall across short, medium, and long posts. One possible reason is that longer posts contain more varied language. In a long post, a few AI-like phrases may be mixed with many ordinary expressions, making character n-gram evidence less concentrated. The transformer can process a larger context window and may be better at recog-

nizing broader writing patterns across the post.

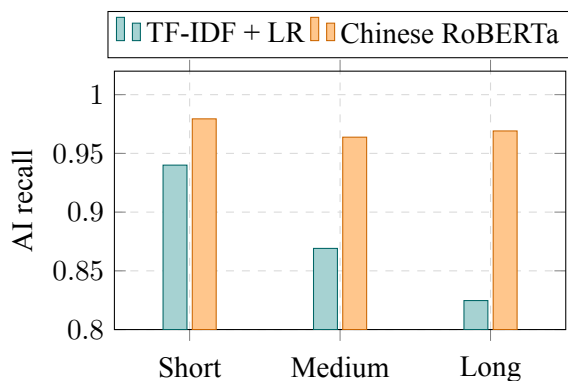


Figure 3: AI recall by text-length bucket on the test split.

At the same time, the transformer has lower AI precision on long posts. This suggests that long human posts may sometimes share features with AI-generated writing, such as organized structure, polished wording, or explanatory tone. This is a critical point for practical use. A detector that performs well on average may still over-flag careful or formal human writing. Therefore, long-post false positives should be manually inspected in future work. This would help determine whether the model is detecting AI generation, or whether it is partly learning a proxy such as polished writing style.

3.3 What Was Not Evaluated

Several components remain unimplemented or only partially represented. The report does not perform semantic analysis, sentiment analysis, discourse analysis, qualitative coding, LLM-as-classifier detection, content-only/title-only ablations, word-segmented TF-IDF ablations, held-out-generator evaluation, or domain-held-out evaluation. Error analysis currently consists of false-positive and false-negative outputs; there is no separate implemented qualitative analysis report or correctness-group text-statistics report.

4 Conclusion

The implemented project completes a focused, reproducible comparison between a text-only TF-IDF baseline and a text-only Chinese RoBERTa-style transformer for detecting AI-generated RedNote

posts. On the fixed held-out test split, the transformer is stronger by macro-F1, AUROC, average precision, log loss, and especially AI recall. The practical tradeoff is that it produces more human false positives at the default threshold, while the TF-IDF baseline remains strong, simple, and more transparent.

The main limitation of this project is that the evidence comes from a fixed supervised split. Although the split is held out and duplicate leakage is checked, the training, validation, and test sets are still drawn from the same dataset construction process. This means the results support in-distribution performance, but they do not prove robustness to future RedNote posts, new prompt styles, new domains, or unseen AI generators. This is especially important because AI-generated text changes quickly as new models and prompting strategies appear.

A second limitation is that the current comparison does not include ablation experiments. The report does not separately test title-only input, content-only input, word-segmented TF-IDF, different n-gram ranges, different transformer checkpoints, or different maximum sequence lengths. These ablations would help explain why each model performs the way it does. For example, a content-only ablation could show whether titles provide meaningful signal, while a word-segmented TF-IDF ablation could test whether character n-grams are actually the best lightweight representation for this dataset.

A third limitation is the lack of qualitative error analysis. The current pipeline exports false-positive and false-negative examples, but the report does not yet manually classify those errors. This matters because numerical metrics cannot explain whether the model is making reasonable mistakes. For false positives, it would be useful to know whether the human posts are unusually polished, template-like, promotional, or advice-oriented. For false negatives, it would be useful to know whether the AI posts are short, informal, emotional, or highly platform-adapted. Such analysis would make the evaluation more interpretable and could reveal whether the detector is relying on undesirable shortcuts.

Future work should therefore focus on robust-

ness and interpretability. A held-out-generator setting would test whether the model can detect AI text from a generator family not seen during training. A domain-held-out setting would test whether the detector transfers to a new RedNote topic area. Threshold tuning should also be performed on the validation set, because different applications may require different balances between AI recall and human precision. Finally, human-centered evaluation is needed before deployment. If a detector is used in moderation, false positives may have social consequences, so the system should be treated as a decision-support tool rather than an automatic final judge.

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